DevOps ISE-I Solutions

2 marks

Module-1

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1. Outline core value leads to devops philosophy

- DevOps encourages close collaboration between development and operations teams.
- It promotes automation of software development lifecycle processes.
- Continuous feedback and improvement are central to DevOps culture.
- It aims for shared responsibility and accountability for software delivery.

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2. State primary goals achieved through devops implementation

- DevOps aims to deliver software faster and more reliably.
- It reduces deployment failures and recovery time.
- It enhances product quality through automation and testing.
- It ensures faster time to market with continuous delivery.

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3. Classify deployment models aligned with devops in cloud environments

- Public cloud allows DevOps teams to deploy on shared infrastructure.
- Private cloud offers better control and security for sensitive data.
- Hybrid cloud supports mixed deployments for flexibility.
- Multi-cloud enables deployment across different providers.

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4. List stages in standard devops pipeline

- Planning defines the scope and goals of the release.
- Development includes coding and building the application.
- Testing validates the functionality and performance of the code.
- Deployment delivers the application to production or staging.

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5. Identify tools commonly used in continuous monitoring

- Prometheus collects and stores real-time metrics.
- Grafana visualizes metrics through dashboards.
- Nagios monitors system health and alerts on failures.
- ELK Stack aggregates and searches log data effectively.

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6. Choose an appropriate automation tool for build phase and justify its fit

- Jenkins automates building and testing of code continuously.
- It integrates with various plugins and version control tools.
- It supports distributed builds across multiple machines.
- Jenkins is open-source and widely adopted in DevOps workflows.

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7. Match devops practice with corresponding agile values

- Continuous integration supports early and frequent delivery.
- Automated testing ensures working software is delivered.
- Frequent releases align with responding to change quickly.
- Cross-functional teams reflect individuals and interactions.

Module-2

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1. Define actions performed in version control branching

- Branching allows developers to work independently on features.
- It helps isolate bug fixes from main codebase.
- Merging integrates branch changes into mainline safely.
- Conflicts are resolved when changes overlap.

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2. Recognise suitable file types managed by version control tools

- Source code files like `.java`, `.py`, `.js` are tracked.
- Configuration files such as `.json` and `.yaml` are versioned.
- Markdown documentation (`.md`) can be version controlled.
- Scripts like `.sh` or `.bat` used in automation are managed.

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3. Name two benefits achieved through automated configuration scripts

- They ensure consistent environment setup across machines.
- Automation reduces chances of manual errors during configuration.
- Time is saved in repetitive setup processes.
- They enable easy scaling and replication of infrastructure.

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4. List major stages involved in continuous integration pipelines

- Code is committed and pushed to a shared repository.
- Automated builds are triggered on every commit.
- Unit and integration tests run after building.
- Artifacts are stored for deployment or release.

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5. Identify matching tools for each devops phase in a pipeline

- Git is used for source code management.
- Maven or Gradle builds and packages the code.
- Selenium performs automated testing.

- Docker and Kubernetes handle deployment.

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6. Classify container orchestration platforms based on feature sets

- Kubernetes provides auto-scaling and rolling updates.
- Docker Swarm offers simpler, native Docker clustering.
- OpenShift includes enterprise tools and CI/CD features.
- Nomad focuses on simplicity and multi-environment support.

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7. Suggest an open source tool used for log aggregation during monitoring

- ELK Stack integrates logs for analysis and visualization.
- Fluentd collects logs and forwards them to various outputs.
- Graylog provides a centralized log management system.
- Loki by Grafana aggregates logs with label-based queries.

Module-3

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1. List two characteristics of distributed version control systems

- Every user has a full copy of the entire repository.
- Changes can be committed locally without internet.
- It supports offline work and later synchronization.
- Branching and merging are fast and efficient.

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2. State two industrial use cases where git enhances code management

- Git enables feature-based branching in agile teams.
- It allows rollback to previous stable versions easily.
- Open-source projects use GitHub for collaboration.

- Enterprises use Git for secure version control at scale.

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3. Identify two key differences between git and cvs

- Git is distributed, whereas CVS is centralized.
- Git allows local commits; CVS requires server access.
- Git supports fast branching and merging.
- CVS lacks modern tooling and performance.

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4. Match git commands with their respective functions in a repository workflow

- `git clone` copies a remote repository to local.
- `git commit` saves staged changes to the local repo.
- `git push` sends local commits to the remote server.
- `git pull` updates local repo with remote changes.

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5. Classify commands under local and remote operations in git

- Local: `git add`, `git commit`, `git status`, `git log`.
- Remote: `git push`, `git pull`, `git fetch`, `git clone`.
- Local operations don’t require internet.
- Remote operations sync with external repositories.

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6. Select branching models suited for collaborative development

- GitFlow organizes branches for features, releases, and hotfixes.
- GitHub Flow supports simple, PR-based workflows.
- Trunk-based development keeps a single shared branch.
- Forking workflow is used in open-source collaboration.

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7. Point out one feature that distinguishes mercurial from git

- Mercurial uses simpler command syntax than Git.

- It emphasizes ease of use over flexibility.
- All commands are consistent and user-friendly.
- It’s built with Python and uses a different storage model.

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5 marks

Module 1

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1. Demonstrate the integration flow from code commit to deployment using DevOps stages

- Developers commit code changes to a shared version control repository like Git.
- The commit triggers a Continuous Integration (CI) pipeline that fetches the latest code.
- Automated build tools like Maven or Jenkins compile the code and package it into deployable artifacts.
- Unit tests and integration tests are run to validate the functionality and ensure no regressions.
- After successful testing, the artifact is stored in an artifact repository like Nexus or JFrog.
- Continuous Deployment (CD) tools such as Spinnaker or ArgoCD deploy the artifact to staging or production environments.
- Configuration files are injected using infrastructure-as-code tools like Ansible or Terraform.
- Post-deployment monitoring is initiated using tools like Prometheus or Datadog to track metrics and logs.
- Alerts are generated for any failures or threshold breaches during runtime.
- Feedback from monitoring is used to trigger rollbacks or fix bugs in the next iteration.

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2. Illustrate typical transitions observed in teams shifting from Agile to DevOps

- Agile teams initially focus on iterative development with frequent releases; DevOps expands this to automate delivery.
- Manual testing in Agile shifts to automated testing in DevOps for faster feedback.
- Separate development and operations teams merge to form cross-functional DevOps teams.
- Continuous Integration evolves into full Continuous Delivery or Deployment pipelines.
- Agile emphasizes planning and sprints, while DevOps focuses on automation and reliability.
- Documentation becomes code-centric with IaC (Infrastructure as Code) practices.
- Testing moves left (early in the cycle) and becomes part of the build process.
- Release cycles change from bi-weekly/monthly to daily or even multiple times a day.
- Post-release monitoring and observability become critical in DevOps.
- Agile principles remain intact, but DevOps complements them with operational efficiency.

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3. Examine the shifts in developers' responsibilities with DevOps adoption

- Developers take ownership of not just coding but also deploying and monitoring their applications.
- They write unit, integration, and even acceptance tests as part of the development process.
- Developers must understand CI/CD pipelines and may configure them with YAML or scripting.
- They contribute to infrastructure using Terraform, Docker, or Kubernetes files.
- Logging, observability, and monitoring become part of the development responsibility.

- Security practices like static code analysis and vulnerability scans are integrated by developers.
- Feature flags and rollback strategies are planned during development.
- Developers participate in incident response and are on-call for production issues.
- Collaboration with QA and operations teams increases significantly.
- Overall, developers shift from “code complete” to “feature delivered and running safely.”

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4. Apply DevOps principles to reduce lead time in release cycle scenario

- Automate the build and test processes to shorten feedback loops using tools like Jenkins or GitHub Actions.
- Implement Continuous Integration so that every code push is validated immediately.
- Use containerization (e.g., Docker) to ensure consistent deployment environments.
- Employ blue-green or canary deployments to safely release new versions.
- Integrate monitoring and alerting systems to detect post-deployment issues quickly.
- Use infrastructure-as-code to provision environments faster and consistently.
- Implement versioned artifacts and rollback mechanisms to ensure safe releases.
- Enable developers to self-deploy to staging using controlled access and audit logs.
- Reduce handoffs between teams by building cross-functional DevOps teams.
- Analyze metrics like change failure rate and MTTR to continuously improve the release pipeline.

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5. Construct a deployment workflow incorporating continuous testing and integration

- Code is pushed to a remote repository such as GitHub or GitLab.
- A CI tool like Jenkins or GitHub Actions triggers a build job automatically.

- The build pipeline includes linting, unit testing, and static code analysis.
- If tests pass, the code is packaged and sent to an artifact repository.
- A separate test stage runs integration and end-to-end tests in an isolated environment.
- On successful testing, the CD pipeline triggers automated deployment to staging.
- Automated smoke and regression tests are executed post-deployment.
- Manual testing is optionally done by QA before production push.
- Once verified, the artifact is promoted and deployed to production.
- Monitoring and alerting tools track performance and raise issues for rollback if needed.

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6. Justify the role of cloud native practices within DevOps environments

- Cloud-native architectures support microservices, enabling independent deployment and scaling.
- DevOps relies on dynamic provisioning, which is enabled by cloud-native tools like Kubernetes.
- Infrastructure-as-Code (IaC) allows consistent, version-controlled infrastructure deployments.
- Auto-scaling and fault tolerance provided by cloud platforms reduce downtime.
- Cloud-native storage and databases integrate smoothly into CI/CD pipelines.
- Observability tools like Prometheus and Fluentd support logging and monitoring in cloud-native systems.
- DevOps teams can deploy across regions using cloud-native load balancing and replication.
- Cloud-native services (e.g., AWS Lambda, Azure Functions) support faster prototyping and reduced ops burden.
- Security is managed via cloud-native IAM, secrets management, and network policies.
- Overall, cloud-native environments complement DevOps goals of automation, scalability, and resilience.

7. Analyze the coordination between DevOps engineers and QA in microservices projects

- DevOps engineers set up CI/CD pipelines that integrate test automation defined by QA.
- QA teams write test cases that run on every commit or pull request via CI.
- Both teams collaborate on test data management and mocking strategies for isolated services.
- DevOps handles deployment environments, while QA ensures they are test-ready.
- QA reports bugs using monitoring tools integrated into the pipeline, enabling faster fixes.
- Shared dashboards help both teams track performance, uptime, and errors.
- Containerized test environments help QA test microservices in production-like conditions.
- Feature flags allow QA to test incomplete features without affecting other services.
- QA ensures non-functional testing (like load or security) is integrated into CD pipelines.
- Close coordination ensures faster feedback, fewer production bugs, and higher release confidence.

Module - 2

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1. Describe the steps in establishing a version control workflow in a team

- Select a version control system such as Git based on project requirements and team familiarity.
- Define a branching strategy like GitFlow or trunk-based development to manage parallel work.

- Set up a central repository (e.g., GitHub, GitLab) with access control and protected branches.
- Implement pull request or merge request workflows for code review and collaboration.
- Establish naming conventions and commit message guidelines for clarity and consistency.
- Integrate CI tools to run tests and enforce quality checks on each pull request.
- Define policies for conflict resolution, rebase vs. merge decisions, and release tagging.
- Schedule regular branch merges or cleanup sessions to maintain repository hygiene.
- Provide onboarding documents and training to align team members with the workflow.
- Continuously improve the workflow based on retrospectives and feedback loops.

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2. Analyse the transition from traditional deployment to continuous methods

- Traditional deployments involved manual steps, downtime, and coordination across teams.
- Continuous methods automate building, testing, and deploying applications through pipelines.
- Teams move from periodic big releases to smaller, incremental updates.
- Manual testing is replaced by automated unit, integration, and regression tests.
- Infrastructure provisioning shifts from manual scripts to Infrastructure as Code.
- CI/CD pipelines become the backbone of deployment, ensuring repeatability.
- Version control plays a critical role in tracking changes and triggering builds.
- Risk is reduced with blue-green or canary deployments in continuous models.
- Monitoring and observability become essential to validate live deployments.
- The transition enables faster feedback, reduced lead time, and improved product quality.

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3. Design a tool chain that connects integration testing and deployment

- Code is committed to GitHub or GitLab as the source control repository.
- Jenkins or GitHub Actions fetches the code and triggers build jobs.
- Testing frameworks like JUnit, Mocha, or Cypress run unit and integration tests.
- Test results are reported in real-time through tools like Allure or SonarQube.
- If tests pass, the artifact is built and stored in Nexus or JFrog Artifactory.
- Deployment tools like ArgoCD or Spinnaker deploy artifacts to staging or production.
- Containerization with Docker ensures consistent environments across testing and deployment.
- Kubernetes manages orchestration of containers across multiple environments.
- Prometheus and Grafana provide observability post-deployment.
- Feedback loops inform developers immediately in case of failure or regression.

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4. Evaluate configuration drift in cloud managed environments

- Configuration drift occurs when actual system state deviates from the intended configuration.
- It often results from manual changes, inconsistent automation scripts, or ad-hoc updates.
- In cloud environments, drift can lead to security vulnerabilities, broken deployments, or performance issues.
- Tools like Terraform detect and correct drift by comparing live infrastructure with declared state.
- Drift tracking is essential in large environments with frequent provisioning changes.
- Periodic drift detection scans help maintain compliance and reliability.
- Cloud-native tools like AWS Config or Azure Policy assist in drift remediation.
- Version controlling infrastructure definitions enables consistent deployment practices.

- Integrating drift detection into CI/CD workflows improves visibility and response time.
- Proper role-based access and automation reduce the chances of untracked manual changes.

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5. Illustrate stages of container lifecycle and isolation benefits

- **Create:** A container image is built from a Dockerfile and stored in a registry.
- **Run:** The container is instantiated from the image and started in an isolated environment.
- **Pause/Stop:** The container process can be temporarily paused or stopped.
- **Restart:** Containers can be restarted after failure or system reboot.
- **Remove:** The container instance is deleted, freeing up resources.
- Isolation is achieved via namespaces for process, network, and file system.
- Control groups (cgroups) limit resource usage like CPU and memory per container.
- Containers run independently, minimizing interference between applications.
- This improves security, debugging, and performance predictability.
- Developers can replicate environments reliably across systems using the same container image.

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6. Choose tools for implementing configuration management in hybrid infrastructure

- **Ansible** is agentless and suitable for automating cloud and on-prem setups.
- **Chef** uses a pull-based model and is ideal for complex, large-scale configurations.
- **Puppet** is widely used in enterprises and integrates well with hybrid environments.
- **Terraform** manages infrastructure as code across cloud providers like AWS, Azure, and GCP.
- **SaltStack** provides fast remote execution and scalable management.

- **Cloud-init** automates initial VM configuration in hybrid deployments.
- **AWS Systems Manager** and **Azure Automation** manage cloud-native configurations.
- Each tool supports idempotent operations to avoid unintended changes.
- YAML or DSL-based configuration files are version-controlled for traceability.
- These tools ensure consistency, compliance, and scalability across environments.

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7. Map each monitoring layer with appropriate feedback matrix

Monitoring Layer	Purpose	Tools	Feedback Metric
Infrastructure Monitoring	Tracks CPU, memory, disk, and network usage	Prometheus, Nagios	Resource utilization, uptime
Application Monitoring	Observes application performance and errors	New Relic, AppDynamics	Response time, error rate
Log Monitoring	Captures logs for audit and debugging	ELK Stack, Fluentd	Log error frequency, exceptions
User Experience Monitoring	Measures real user interaction	Google Lighthouse, Dynatrace	Load time, UI responsiveness
Security Monitoring	Detects vulnerabilities and breaches	OSSEC, Falco, Wazuh	Intrusion attempts, access logs

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Module - 3

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1. Describe Git’s role in modern software development pipelines

- Git acts as the backbone of version control in most CI/CD pipelines.

- It enables developers to track changes, manage branches, and collaborate on code.
- Git repositories trigger CI tools like Jenkins or GitHub Actions upon code push.
- Branching in Git supports parallel feature development and testing workflows.
- Git history helps in auditing, bug tracking, and reverting problematic code.
- It integrates with cloud platforms like GitHub, GitLab, and Bitbucket.
- Pull requests in Git allow structured code reviews and team approvals.
- Tags and release branches facilitate controlled versioning and deployment.
- Hooks in Git automate tasks like linting or testing before a commit is accepted.
- Git ensures a structured, traceable, and collaborative development lifecycle.

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2. Compare CVCS and DVCS in terms of scalability and collaboration

- DVCS like Git scales better since every user has a complete local copy of the repository.
- CVCS requires constant server communication, limiting offline work and scalability.
- In DVCS, collaboration is enhanced through features like pull requests and branch merging.
- CVCS follows a linear workflow, making concurrent development more error-prone.
- DVCS supports distributed teams by allowing asynchronous work and syncing.
- CVCS performance degrades with large projects and many contributors.
- DVCS offers better conflict resolution tools and commit history management.
- Overall, DVCS is more robust and flexible for modern, collaborative development.

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3. Demonstrate the usage of Git commands to clone, commit, push and pull from a repository

- `git clone <repo_url>` is used to create a local copy of a remote repository.

- After changes, `git add <file>` stages the changes for commit.
- `git commit -m "message"` records the staged changes to the local repository.
- `git push origin <branch>` uploads the local commits to the remote repository.
- `git pull origin <branch>` fetches and merges remote changes into the local branch.
- These operations form the core cycle of Git-based collaborative development.
- Git also supports SSH or HTTPS for secure remote access.
- Proper usage ensures smooth synchronization and teamwork.
- Conflicts, if any, are resolved manually before successful push/pull.
- These commands are platform-independent and work in all Git hosting platforms.

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4. Analyse a use case involving Git branching strategy in a feature release cycle

- In a feature release cycle, developers create a new branch from `main` or `develop`.
- They isolate their work using branches named like `feature/login-auth`.
- Code is committed and pushed to the feature branch regularly.
- Once complete, a pull request is opened for code review and CI testing.
- After approval, the feature branch is merged back into `develop`.
- The `release` branch is created from `develop` for testing and staging.
- Bugs in release are fixed directly in the `release` branch and merged to `main` and `develop`.
- After testing, the `release` branch is merged into `main` and deployed.
- A `hotfix` branch can be used if critical bugs are found in `main` post-release.
- This strategy allows isolated development, testing, and stable production releases.

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5. Organize a table comparing Git workflows like Git Flow, GitHub Flow, and Forking Workflow

Aspect	Git Flow	GitHub Flow	Forking Workflow
Branches Used	`main`, `develop`, `feature`, `release`, `hotfix`	`main`, `feature`	Forked repo, `main`, `feature`
Use Case	Enterprise projects with long release cycles	Continuous delivery, web apps	Open-source or external contributions
Collaboration Model	Centralized, team-based	Centralized, team-based	Decentralized, individual contributions
Complexity	High (many branches)	Low (simpler model)	Medium (involves forks and PRs)
CI/CD Integration	Good	Excellent	Good

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6. Illustrate the sequence of the Git operations involved in working with remote repositories

- First, clone the repository using `git clone <url>` to create a local copy.
- Create a new branch using `git checkout -b <branch>` for feature development.
- Make code changes and use `git add .` to stage all modified files.
- Commit the changes with `git commit -m "your message"`.
- Push the branch using `git push origin <branch>` to the remote.
- Open a pull request from the branch to the main repository branch.
- Review and merge the changes via the platform or CLI tools.
- Periodically use `git pull origin <branch>` to sync local with remote.
- Resolve merge conflicts if they arise during the pull or merge.

- Use `git fetch` to get updates without merging, useful for checking upstream status.

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7. Distinguish Git from Mercurial based on branching, ease of use and platform integration

- **Branching:** Git supports lightweight, fast branching using references; Mercurial uses named branches and bookmarks.
- **Ease of Use:** Mercurial offers a simpler and more consistent CLI; Git is more flexible but has a steeper learning curve.
- **Platform Integration:** Git is supported natively on GitHub, GitLab, Bitbucket; Mercurial support has declined in mainstream platforms.
- Git is preferred for open-source and enterprise projects due to its robust ecosystem.
- Mercurial is still used in some legacy or internal projects (e.g., Mozilla's older infrastructure).
- Git has richer tooling for visualization, branching strategies, and integrations.
- Mercurial simplifies workflows by reducing command inconsistencies found in Git.
- Git supports more complex workflows like rebase, squash, and cherry-pick.
- Overall, Git is more widely adopted, while Mercurial focuses on simplicity and ease of learning.
